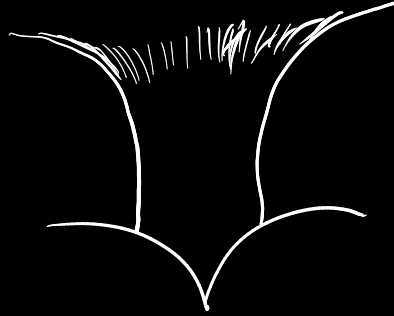


Agentic AI for Knowledge Graph Reconstruction for assisting Forensic Investigations

We will be taking different datas as inputs, like

- ① FIR reports
- ② CCTV footage
- ③ Call Detail Record (CDR)
- ④ GPS/Location tracking
- ⑤ Bank Transactions
- ⑥ Social Media / Chat Data
- ⑦ Vehicle / License Plate
- ⑧ Audio Recordings



→ All this different data types will go through their own preprocessing

→ After preprocessing, we'll create nodes and edges from each preprocessing as json

After Preprocessing,

① nodes
edges

② nodes
edges

③ nodes
edges

④ nodes
edges

⑤ nodes
edges

⑥ nodes
edges

⑦ nodes
edges

We will add all the pipelines into "graph integration layer"

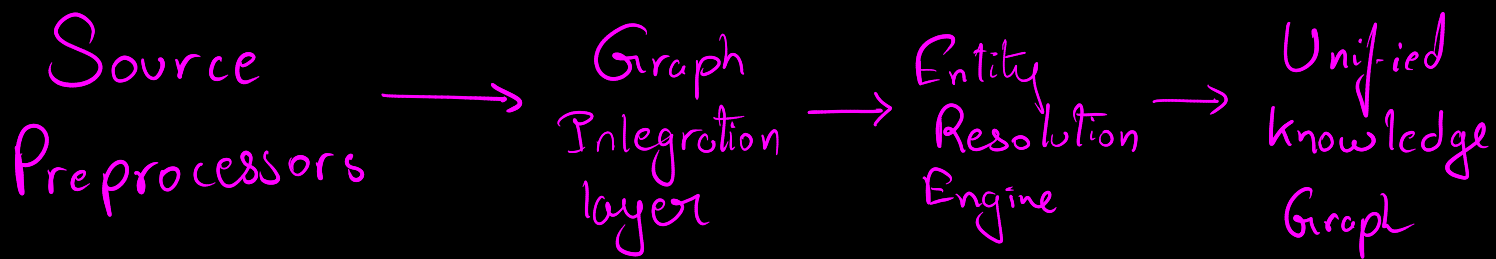
Then, we will pass it through "Entity Resolution Layer"

→ identity Resolution
→ entity Matching
→ deduplication

→ merges duplicate entities
→ resolves identities
→ updates relationships
→ handles confidence scores

This layer does this

Final Pipeline Till This :-



Knowledge Graph → we can show it in dashboard now.

↳ Will pass it through GNN & Rule validation layer

Before passing it to GNN,

① Graph Feature Engineering

Knowledge graph is basically

- nodes
- edges
- timestamps
- relationships

→ node feature generation :-

↳ Every nodes receives a feature vector.

↳ Extracted feature could be

- Call Frequency
- Unique contacts
- Night movement ratio

↳ Example vector :-

Rahul = [52, 18, 0.81, 50000, 0.74, 0.62]

- Transaction volume
- Crime scene proximity
- Community score

→ Edge features :-

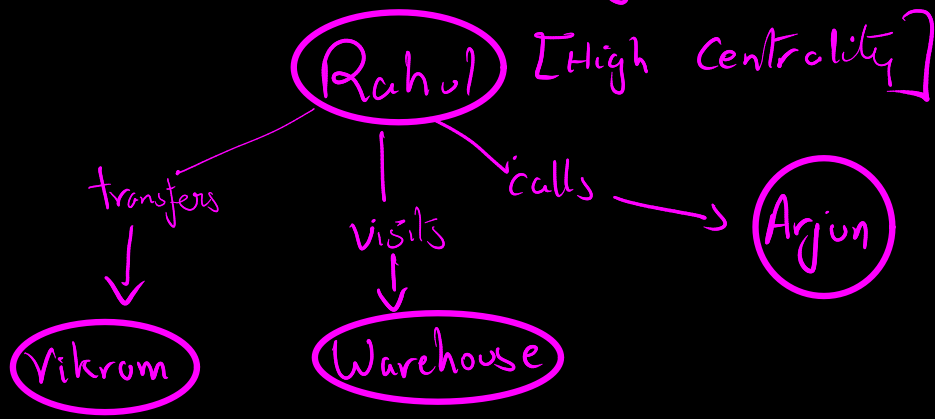
↳ features like

- call count
- Avg duration
- Recency score
- Temporal Anomaly

→ Structural Graph Features

↳ degree centrality, betweenness centrality, clustering coefficients, Pagerank, shortest paths, graph motifs

After this, Featured graph is ready →



② Graph Analytics layer

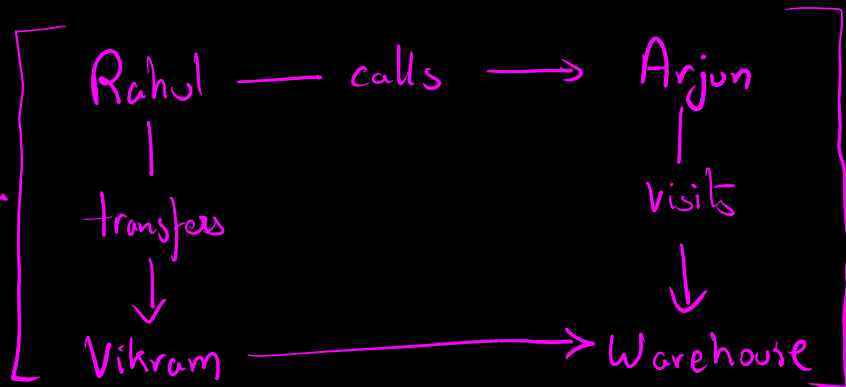
↳ Perform classical graph analytics before going to GNN
* Many useful insights come directly from graph theory.

Classical Graph Operations →

- a/ Centrality Analysis: Detect highly influential nodes
 - b/ Community Detection: Detect Clusters / groups
 - c/ Path analysis: Find hidden indirect connections
- can show it in UI with colored groups...

Visualisation :-

Criminal Clusters



Community Detection

③ GNN Intelligence Layer

"Graph Neural Network" learns relational behavior patterns

GNN Tasks :-

a/ Link Prediction — Predict hidden relationships

Rahul — calls —> Arjun

Arjun — visits —> Warehouse

GNN Predicts :- Rahul likely associated with Warehouse
Confidence: 0.87

b/ Suspicion Scoring — classify high-risk entities

GNN Predicts :- { "entity": "Rahul", "risk-scoring": 0.92 }

c/ Anomaly Detection — Detect unusual patterns.

↳ Sudden communication spikes

↳ Synchronised movements

↳ abnormal transaction behaviour

Visualisations :-

Rahul — calls —> Arjun
Rahul - - - - likely visits - - - -> Warehouse

Solid Edge → Observed Evidence
Dotted Edge → GNN Inference

④ Rule-based Validation layer → GNN prediction are probabilistic

Rules validate logical consistency.

Example Rule:-

IF: suspects near crime scene
communication spike before event
transaction after event

THEN: increase suspicion score

Why Rahul flagged? Visualisation

- ✓ 14 calls before accident
- ✓ GPS near warehouse
- ✓ Transaction after robbery
- ✓ connected to known suspect

⑤ Temporal Reconstruction Engine

The system reconstructs event sequences

Example Timeline:-

8:45 PM: Rahul leaves home

9:10 PM: Calls Arjun

9:30 PM: near warehouse

10:00 PM: transaction occurs

Temporal Insights

↳ System detects:

⇒ coordination

⇒ preparation

⇒ post event activity

⇒ escape pattern

Visualisation:-

8:45 PM	Rahul leaves home
9:10 PM	Rahul Calls Arjun
9:30 PM	Rahul near warehouse
10:00 PM	Money transfer

⑥ Behavioral & Anomaly Analysis

↳ System analysis: movement patterns
communication patterns
behavioral deviations

Example:- Rahul usually make 2 calls/day
Today: 27 calls before robbery

Behavioral Features:-

- unusual travel
- sudden financial activity
- synchronised movements
- repeated co-location

Visualisation:-

Movement heat maps

↳ Crime hotspot intensity map

Communication spike

↳ Call activity vs time graph

⑦ Investigation Reasoning Engine

All intelligent layer combines :-

Inputs :- → graph analytics
→ GNN outputs
→ rule engine
→ timelines
→ anomaly scores

This engine synthesizes :-
"investigative hypothesis"

Example Output :-

Possible coordinated robbery network detected.

Entities:

- Rahul
- Arjun
- Vikram

Supporting Evidence:

- communication spike
- shared location overlap
- suspicious transfer
- temporal synchronization

Confidence: 0.82

⑧ Explainability Layer

Every AI insight must be explainable

↳ System explains → why a relation exists
↳ what evidence supports it
↳ which models contributed
↳ confidence scores

Prediction scores :-

GNN link pred :- 0.84

GPS proximity :- 0.91

Communication corr :- 0.79

Visualisation :- Clicking an edge shows →

Rahul → Warehouse

Evidence:

- GPS overlap
- indirect network path
- temporal consistency

Generated by:

- GAT model
- rule validation layer

⑨ Investigator Dashboard

Interactive Graph → relationship exploration

Timeline View → event reconstruction

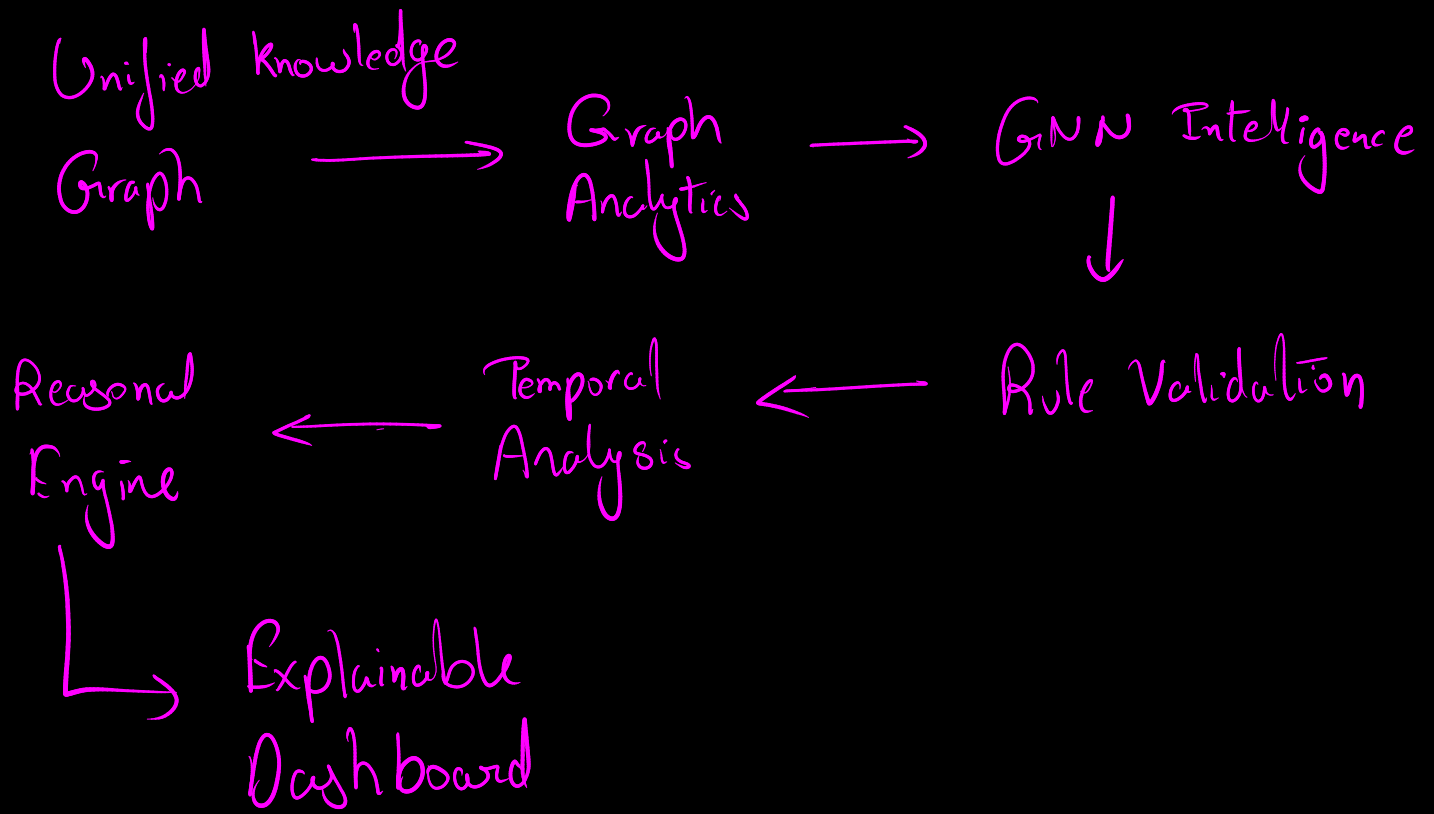
Heat maps → spatial analysis

Suspicion Ranking → priority investigation

Query Engine → graph search

Evidence Explorer → traceability

FINAL SYSTEM OVERFLOW



Btw, Rahul
is innocent

